Predictive Multitask Learning for Streaming Code-Switching

CS6140 Machine Learning — Project Update 1 Report

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Abstract

We study anticipatory code-switching: after reading the prefix $x_1, \ldots, x_t$, predict whether the next word x_{t+1} switches language and how long the new-language span will last. This report describes the data pipeline we built for all 15 language pairs of SwitchLingua — language-specific word segmentation, word-level language identification, switch and duration labels, a deduplicated 70/10/20 split, and a streaming loader whose inputs never include x_{t+1} — and our first baselines. A naive model that only sees the current word reaches a mean switch-class F1 of .426 across pairs (SD .102), roughly twice a model that only knows the current word's language. Code: https://github.com/StarfishJ/CS6140ML_Project.

1 Problem Description

Code-switching is the alternation between two or more languages within an utterance or conversation. Most NLP work recognizes a switch after the mixed-language text has been observed. We instead study **anticipatory** code-switching: at word x_t, using only the prefix $x_1, \ldots, x_t$, predict (1) the switch label $y_{sw}(t) = 1$ if x_{t+1} is in a different language than x_t, and (2) the duration label $y_{dur}(t)$, the length of the new-language span starting at x_{t+1}: 0 for 1–2 words, 1 for 3–6, 2 for 7 or more, and -1 (ignored by the loss) when there is no switch. Real-time systems such as bilingual predictive keyboards, dialogue agents and streaming speech recognition need to know a switch is coming before it happens. Our central question is **universality**: does one method anticipate switches equally well across typologically different pairs? We measure it with the same pipeline and metric on every pair, reporting per-pair switch-class F1 ("Anticipatory F1") and its standard deviation across pairs. Our final model will be a multilingual transformer (XLM-R or mBERT) with a causal attention mask and two heads, one for y_{sw} and one for y_{dur}.

2 Background and Related Work

Code-switch prediction. Solorio and Liu [11] first framed the prediction of code-switching points in Spanish–English conversation as a learning task. They trained Naive Bayes and VFI classifiers on lexical, part-of-speech and language-ID features of the preceding words. Results were encouraging, but the feature-based approach was tested on one small corpus and one pair. Ostapenko et al. [8] predicted code-switching points in English–Spanish dialogue. They truncated each input at a candidate point so the model sees only the prefix, and added sociolinguistic speaker features as prompts. Speaker information improved accuracy, but evaluation covered one pair and no switch duration. Winata et al. [13] built code-switched language models for English–Mandarin. A pointer-generator network learned where to switch between aligned parallel sentences and produced synthetic code-switched training data. Language-model perplexity improved, but switching was modeled implicitly through the language model, on one pair.

Language identification in code-switched text. Solorio et al. [10] ran the first shared task on word-level language identification in code-switched social-media text, for Spanish–English, Nepali–English, Mandarin–English and Modern Standard Arabic–dialectal Arabic. Participants used CRFs and other supervised sequence models with character and lexical features. Accuracy was high for some pairs but much lower for the closely related Arabic varieties. Molina et al. [6] organized the second edition for Spanish–English and Modern Standard Arabic–Egyptian Arabic tweets. Systems ranged from CRFs to neural networks with word and character features. Spanish–English was again much easier than the Arabic pair, showing that closely related languages remain hard. Aguilar et al. [1] released LinCE, a benchmark of ten corpora over four pairs (including Hindi–English and Spanish–English) and four tasks (language ID, POS tagging, NER, sentiment). They reported baselines from BiLSTMs to multilingual BERT. Language ID is close to solved with full-sentence context, but all tasks look at text after it is written. Khanuja et al. [4] proposed GLUECoS, a code-switched NLP benchmark for English–Hindi and English–Spanish covering language ID, POS, NER, sentiment, question answering and NLI. They fine-tuned mBERT and also pre-trained it on synthetic code-switched text. The extra pre-training helped, but the benchmark covers only two pairs and no anticipatory task.

Multilingual encoders. Devlin et al. [3] introduced BERT, a Transformer encoder pre-trained with masked language modeling and next-sentence prediction. Fine-tuning it set new results on eleven tasks including GLUE and SQuAD. Its attention is bidirectional, so prefix-only prediction requires a causal mask. Pires et al. [9] probed multilingual BERT. Zero-shot transfer experiments showed that mBERT transfers across languages and can process code-switched and transliterated input. Transfer drops for typologically distant pairs and for transliterated text, which motivates our per-pair variance analysis. Conneau et al. [2] trained XLM-R, a masked language model on 2.5 TB of filtered CommonCrawl text in 100 languages. It outperformed mBERT on cross-lingual classification, QA and NER, especially for low-resource languages. It is our strongest candidate backbone, although, like BERT, it is bidirectional. Muller et al. [7] studied how multilingual models handle languages unseen in pre-training. They found large differences between languages and that transliterating a language into the script of a related high-resource language helps. This supports treating romanized Hindi with a model trained on romanized text.

Multitask learning and data. Liu et al. [5] proposed MT-DNN, which shares a BERT encoder across tasks with task-specific output layers trained jointly. It improved GLUE results, but all tasks were

sentence-level classification over fully observed input, unlike our prefix-only switch and duration heads. Xie et al. [14] introduced SwitchLingua, generated with the LinguaMaster multi-agent framework under linguistic constraints, with 420K text samples across 12 languages and an audio part. It is synthetic, but it is the only resource with the breadth of pairs a universality study needs.

3 Dataset

Source. We use the text part of SwitchLingua (SwitchLingua_text on Hugging Face): 15 pairs, 234,172 rows, 14 X–English pairs plus German–French. Each row is a generated conversation whose `data_generation_result` field is a list of utterances, with metadata such as topic and switch type. There are no word-level language labels, so we derive them.

Segmentation and word language ID. Word units and language ID depend on the writing system (Table 1). Space-separated languages are split on whitespace and punctuation, keeping combining marks (an early version of our regex split 44.5% of Devanagari words in the first 1,000 Hindi rows into fragments). Chinese, Cantonese and Japanese are segmented with jieba, PyCantonese and GiNZA bunsetsu units (a content word plus its particles, comparable to the Korean eojeol); Vietnamese syllables are merged into words by pyvi. A word mixing Latin and another script (Korean "presentation-eseo") is split at the script boundary, and a hyphenated word whose parts are clearly in different languages ("nag-meeting") is split at the hyphen. For non-Latin pairs the Unicode script gives the language. Hindi utterances containing Devanagari use script as well; the remaining 59.5% of Hindi rows are fully romanized and are tagged with `muril-hinglish-lid`, a MuRIL model fine-tuned on LinCE Hindi–English. For the eight Latin-script pairs, a word gets the language in which it is clearly more frequent according to wordfreq [12] (Zipf difference ≥1.0, or 0.5 toward English for Malay and Filipino, whose lists contain many English loans), or the language indicated by letters only one side uses. Ambiguous words take the language of the nearest decided words on both sides. Numbers inherit the previous word's language. On spot-checked validation samples (about 400–500 words per Latin-script pair), word-level agreement with our own reading was about 98–99%; these are weak labels, not gold annotations.

Labels and split. From the word language IDs we compute y_{sw} and y_{dur} for every position whose two words have a known language (unknown words are 0.0–0.2% of words, mostly utterance-initial numbers). Exact duplicate rows are removed, rows that share any utterance are grouped, and groups are split 70/10/20 into train, validation and test with a fixed seed per pair; 0 utterances appear in more than one split. Cantonese–English (119,248 rows, 2,652 exact duplicates) is downsampled after splitting to 8,381 rows, the median size of the other pairs.

Streaming loader and causal inputs. `StreamingCSDataset` is a PyTorch `IterableDataset`: a generator reads one CSV row at a time and yields one example per word t — the window $x_{t-31}, \ldots, x_t$, the language of x_t, y_{sw} and y_{dur}, and optionally XLM-R/mBERT subword ids scored at the last subword of x_t. Nothing is precomputed. The last word of an utterance yields no example and windows never cross utterances. Labels may read the whole sentence, since they

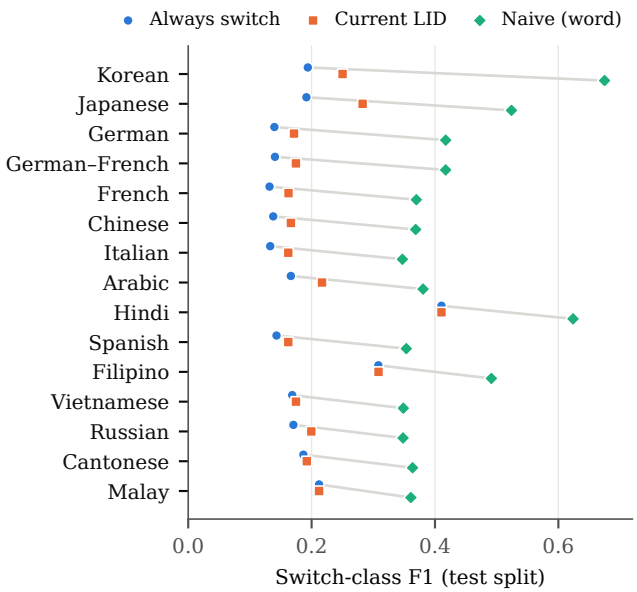


Figure 1: Test switch-class F1 per pair (values in Table 2). Pairs are ordered by the naive model's gain over always switch; the grey bar joins always switch and naive.

are the answers, but inputs never include x_{t+1} in training or testing: the text passed to a tokenizer ends at x_t, and the input LID is recomputed from the prefix only (left context for Latin-script pairs, the Hinglish model run on each prefix for romanized Hindi). Re-running the pipeline on the prefix alone for 40 validation utterances per pair gives an identical LID input in 14 pairs (Vietnamese: 99.9%) and an identical word x_t in 12 pairs; Cantonese, Japanese and Vietnamese match at over 99% because their segmenters look at following characters.

4 Experimentation and Results

Setup. All scores are switch-class F1 on the test split (65K–110K word positions per pair). Always stay never predicts a switch (F1 = 0 on every pair). Always switch predicts a switch at every word; its F1 depends only on the switch rate p, $2p/(1+p)$, and serves as the reference line. Current LID estimates $P(\text{switch} \mid \text{pair}, \text{LID of } x_t)$ on train with add-one smoothing. The naive baseline estimates $P(\text{switch} \mid \text{pair}, x_t)$ for the word itself, shrunk toward its language's rate with five pseudo-counts. Both use only x_t (no punctuation after it, no earlier words), and their thresholds are tuned on each pair's validation split. Gain is naive F1 minus always-switch F1.

Results. Table 2 and Figure 1 show that knowing the word itself, not only its language, roughly doubles F1: from .216 to .426 on average. The current language alone helps little: it beats always switch by at most .091 and not at all for Hindi, Filipino and Malay, where switching is frequent and goes both ways. The naive model gains most for Korean–English (+.481): Korean words with the sentence-final endings *-yo* / *-da* are followed by a switch 97% / 87% of the time, against 11% for other Korean words. It gains least for Malay–English (+.149). Across pairs the naive F1 has SD .102, so

Table 1: The 15 SwitchLingua pairs after preprocessing. Positions are word positions with a label (all splits); S/M/L is the share of switches whose new span has 1–2 / 3–6 / 7+ words; Unk. is the share of words with no language.

Pair	Segmentation	Word LID	Rows	Kept rows	Positions	Switch %	S/M/L %	Unk. %
Arabic–English	space	script	8,548	8,543	359K	9.1	41/24/35	0.00
Hindi–English	space	script+MuRIL	8,381	8,380	444K	26.0	67/22/12	0.02
Korean–English	eojeol	script	8,266	8,266	347K	10.6	40/22/38	0.01
Russian–English	space	script	8,017	8,017	352K	9.3	40/26/34	0.00
Chinese–English	jieba	script	8,458	8,457	534K	7.5	35/21/44	0.03
Cantonese–English	PyCantonese	script	119,248	8,381	549K	10.3	46/30/24	0.02
Japanese–English	GiNZA	script	8,385	8,384	369K	10.6	37/21/41	0.15
Spanish–English	space	wordfreq	8,110	8,101	421K	7.7	35/26/39	0.02
French–English	space	wordfreq	8,070	8,067	399K	7.1	34/24/42	0.03
German–English	space	wordfreq	6,819	6,814	312K	7.7	37/23/40	0.04
Italian–English	space	wordfreq	7,979	7,978	405K	7.1	34/24/43	0.03
Malay–English	space	wordfreq	8,386	8,386	415K	11.9	47/26/27	0.11
Filipino–English	space	wordfreq	8,537	8,534	451K	18.0	56/26/18	0.18
Vietnamese–English	pyvi	wordfreq	8,457	8,455	463K	9.3	43/24/33	0.03
German–French	space	wordfreq	8,511	8,506	378K	7.6	32/26/42	0.04
Total			234,172	123,269	6.20M			

Table 2: Test switch-class F1, sorted by gain of the naive (current-word) baseline over always switch.

Pair	Always sw.	Cur. LID	Naive	Gain
Korean–English	.194	.250	**.675**	+.481
Japanese–English	.192	.283	**.524**	+.332
German–English	.140	.172	**.417**	+.278
German–French	.141	.175	**.417**	+.277
French–English	.132	.163	**.370**	+.238
Chinese–English	.138	.167	**.369**	+.231
Italian–English	.133	.162	**.347**	+.214
Arabic–English	.166	.217	**.381**	+.214
Hindi–English	.410	.410	**.624**	+.213
Spanish–English	.143	.162	**.353**	+.210
Filipino–English	.308	.308	**.491**	+.183
Vietnamese–English	.169	.175	**.349**	+.180
Russian–English	.171	.200	**.348**	+.178
Cantonese–English	.187	.192	**.364**	+.177
Malay–English	.212	.212	**.361**	+.149
Mean (SD)		.216 (.068)	**.426** (.102)	

a single current word is far from universal; the spread is a first measurement of the universality gap a context model must close.

Segmentation unit. The word unit decides how many predictions are scored. Re-running the Japanese pair with three units (Table 3) shows that fine morphemes add many easy no-switch positions: the switch rate falls and current-LID F1 halves, while the naive model, which sees the word itself, loses much less. We use bunsetsu units, comparable to Korean eojeol, so that pairs are scored on similar units.

Subwords and punctuation. With mBERT, 72% of Korean words are split into several subwords (English: 15%), and with XLM-R a piece crosses into the next word for 7.6% of Chinese and 5.1% of Japanese words. We therefore keep labels at the word level, score each word at its last subword, and will tokenize word by word. Inputs exclude the punctuation after x_t: in SwitchLingua a sentence-final mark is followed by a switch 51–93% of the time and precedes 4–33% of all switches, so models that see it score much higher on these inter-sentential switches.

Table 3: Japanese–English with three segmentation units (all splits / test).

Unit	Positions	Switch %	Cur. LID	Naive
Morphemes (fugashi)	624K	6.2	.130	.489
Sudachi mode C	610K	6.4	.134	.489
Bunsetsu (GiNZA)	369K	10.6	.283	.524

5 Plan for Next

- **Causal dual-head model (Update 2).** XLM-R and mBERT with a causal attention mask, switch and duration heads on h_t, loss $\lambda_1 L_{sw} + \lambda_2 L_{dur}$ with `ignore_index=` -1; check against the prefix-by-prefix loader, which must give identical predictions.
- **Stronger baselines.** Previous plus current word, logistic regression on prefix features, a duration-majority baseline, and a variant that sees punctuation, reported separately.
- **Label quality.** A small hand-checked gold set per pair to measure word-LID and switch-boundary accuracy, and incremental segmentation of the prefix for Chinese, Japanese and Vietnamese.
- **Analysis.** Per-pair F1 bar chart and SD for each backbone, duration accuracy at true switches, and inter- versus intra-sentential switches using SwitchLingua's switch-type field; a streaming demo at the end.

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